

DR. MIRIAM KEPPLER

Astronomer • Data Scientist • Space Science & Earth Observation enthusiast

✉ keppler@arizona.edu

📍 Tucson/AZ, USA

🌐 <https://kepplerm.github.io/>

in kepplerm

🔗 kepplerm

EXPERIENCE

Postdoctoral Researcher

The University of Arizona

📅 06/2022 – Ongoing

📍 Tucson/AZ, USA

- Developing a data processing pipeline for the NASA-funded Astrophysics UV SmallSat mission *Aspera*
- Leading the development of an instrument simulator to generate test data to verify the *Aspera* data processing pipeline
- Leading the *Aspera* data management: Defining, developing, and prototyping data products; interfacing with the Archive
- Developing a tool to predict UV geocoronal background line emission from archival data taken by the Far Ultraviolet Spectroscopic Explorer (FUSE) mission
- Successfully lead a proposal for observing time at the highly competitive Hubble Space Telescope (worth 30 orbits of observing time)

Postdoctoral Researcher (Graduate researcher until 11/2020)

Max Planck Institute for Astronomy

📅 11/2016 – 10/2021

📍 Heidelberg, Germany

- Processed and analysed multi-wavelength astronomical data from the near-infrared to sub-millimeter regime
- Led the discovery of the first planet outside the solar system that is still at the stage of formation ([link to press release](#)).
- Published $\gtrsim 40$ refereed articles as first or co-author in international journals
- Coordinated projects within large international teams with 100+ collaborators
- Conceived, proposed and executed observing campaigns as project leader at international telescopes (81 hours of observing time granted)
- Presented oral and written contributions to 12 international conferences
- Communicated research to non-scientific audience

Undergraduate Researcher (Master Project)

Institute of Planetology and Astrophysics Grenoble

📅 10/2015 – 10/2016

📍 Grenoble, France

- Processed and analysed ground-based spectroscopic data to search for new planets
- Executed observing campaigns at telescopes in France and Chile during 18 nights

Undergraduate Researcher (Bachelor Project)

Max Planck Institute for Extraterrestrial Physics

📅 03/2013 – 07/2013

📍 Munich, Germany

- Conceived numerical simulations to model the movement stars in the Galactic Center

ABOUT

PhD astronomer with 8+ years experience in the exploitation and analysis of large and complex data sets in fundamental research. Strong team player with enthusiasm for contributing expertise in an innovation oriented dynamic environment.

STRENGTHS

Problem solving

Communication

Collaboration

Project management

Leadership

Operations

Enthusiasm

Python

Data Analysis

Image processing

Spectroscopy

Scientific programming

Remote sensing

ACHIEVEMENTS



Successful HST observing grant (08/2023)

Got 30 orbits of observing time granted on highly competitive Hubble Space Telescope, including a grant worth \$107k



ESA Astronaut Program finalist (06/2022)

Among top 2% of > 22000 candidates applying for the new corps of Astronauts of the European Space Agency



Otto Hahn Medal (06/2022)

Awarded for an outstanding PhD Thesis by the renowned Max Planck Society



Best PhD Thesis Award (06/2021)

Awarded by Astronomy & Astrophysics, one of the leading international scientific journals of the field



Ernst Patzer Award (11/2018)

Awarded for the best scientific publication by a young astrophysicist in Heidelberg by the Ernst Patzer Foundation (11/2018)

EDUCATION

Ph.D. Astronomy ('summa cum laude')

University of Heidelberg

📅 Nov 2016 – Nov 2020

Thesis: Giant Planet Formation in Disks: An Observational Perspective

Master of Science (M.Sc.) Physics

University of Heidelberg

📅 Oct 2014 – Oct 2016

Thesis: The Search for Extrasolar Planets around Young, Nearby Stars

ERASMUS study abroad, Physics

Université Grenoble Alpes, France

📅 Sept 2013 – Sept 2014

Classes in Physics and Astronomy

Bachelor of Science (B.Sc.) Physics

University of Heidelberg

📅 Oct 2010 – Oct 2013

Thesis: The Galactic Center Cloud G2 and Scattering Processes in the Young Stellar Ring

FURTHER TRAINING

Foundations of Open Science Skills (Fall 2023)

CyVerse / University of Arizona

Learnt about foundations of open science during 8-week long interactive workshop. Topics covered include FAIR data, data management plans, documentation, version control, reproducibility

PI Launchpad Workshop (07/2023)

NASA / University of Michigan

Competitive workshop about how to become a NASA PI.

Leading for Tomorrow Program (11/2021)

German Physical Society (DPG)

Completed competitive program for early career physicists on the training of leadership skills.

COMMUNICATION & ORGANIZATION

- Organizing social events for the University of Arizona (UA) Postdoc Community as part of the UA Postdoc Association.
- Co-organizing ResBaz 2024, a yearly data science conference held at the University of Arizona
- Communicated work to non-scientific audience (articles for amateur astronomy magazine, public talks, interviews, podcasts; e.g., NPR, BBC News). ([link to public talk \[in German\]](#))

TECHNICAL SKILLS

Operating systems: Linux MacOS

Programming: Python basics of C/C++

Tools & libraries: git NumPy SciPy
Pandas Xarray Scikit-learn Plotly
PyCharm

Software: LaTeX Keynote MS Office

LANGUAGES

German (native) ● ● ● ● ●

English (near-native) ● ● ● ● ●

French (near-native) ● ● ● ● ●

Spanish (intermediate) ● ● ● ● ●

Japanese (beginner) ● ● ● ● ●

VOLUNTEERING

🏠 Served as reviewer for the University of Arizona Postdoc travel grants

🏠 Organizing events for postdocs as member of social events committee of University of Arizona Postdoc Association

🏠 Volunteered at the Arizona-Sonoran Desert Museum (Tucson, Arizona)

🏠 Mentored a female high-school student within the STEM platform 'CyberMentor'

MEMBERSHIPS

🏠 German Alpine Club (DAV)

🏠 German Physical Association (DPG)

🏠 Astronomers for Planet Earth (A4E)

HOBBIES

🏠 Playing piano

🏠 Climbing